

Question number	Answer	Mark
2(a)(i)	The only correct answer is D <i>A is incorrect because NADP is not oxidised</i> <i>B is incorrect because reduced NADP is not produced in cyclic photophosphorylation</i> <i>C is incorrect because the NADP is not oxidised</i>	(1)

Question number	Answer	Additional guidance	Mark
2(a)(ii)	C is from {carbon dioxide / CO₂} H is from {water / H₂O} O is from {carbon dioxide / CO₂}	One or two correct = 1 mark All three correct = 2 marks DO NOT ACCEPT any other molecule or incorrect formula DO NOT ACCEPT any other molecule or incorrect formula DO NOT ACCEPT any other molecule or incorrect formula	(2)

Question number	Answer	Additional guidance	Mark
2(b)(i)	An answer that includes three of the following points: • (overall / after 20 °C) rate of increase in C3 plants is lower (1) • the optimum temperature is lower for C3 plants (than C4 plants) (1) • at the optimum temperature, the rate of photosynthesis is slower in C3 plants (1) • credit appropriate comment about rate below 20 °C (1)	NB all mark points are comparative ACCEPT converse throughout for C4 ACCEPT above 20°C C3 plants have a lower rate of photosynthesis NB if values are given they must be correct C3 {26 / 27 / 28} and C4 {34 / 35 / 36} ACCEPT the {fastest / peak} rate for C3 is lower (than the fastest rate for C4) IGNORE any values given e.g. C3 have a higher rate of photosynthesis rate increase in C3 is non-linear but in C4 plants it is linear rate increase in C3 is less steep C3 plants (probably) photosynthesise at	

Question number	Answer	Additional guidance	Mark
2(b)(ii)	- values for R_{20} and R_{10} read from the graph (1) Q_{10} given as 4 (1)	6 and {21.5 / 22} DO NOT ACCEPT with units IGNORE a.u. Bald answer of 4 = 2 marks Bald answer between 3.583 and 3.667 = 1 mark	(2)